1.2 Parent Functions and Transformations

Name _____ Period ____ Date _____

Algebra II 1.2 · Student sheet

I can…

- identify the family of a function.
- describe translations, reflections, stretches, and shrinks.
- describe combinations of transformations and find the domain and range.

Vocabulary · fill in the missing word

Parent function — the _____ function in a family (constant, linear, absolute value, quadratic, …)

Transformation — changes the size, shape, position, or _____ of a graph

Translation — _____ a graph left, right, up, or down

Reflection — _____ a graph over a line

Vertical stretch / shrink — _____ every y-value by a > 1 / by 0 < a < 1

Part 1 · Parent function gallery

Graph y = |x| in Desmos. Then graph each function and compare it with y = |x|.

Function	Vertex	What changed?
y = |x| + 2		
y = |x − 3|		
y = −|x|		
y = 2|x|		
y = ½|x|		

Part 2 · Notes: fill in the blanks

f(x) + k moves the graph _____; f(x − h) moves it _____.

−f(x) is a reflection in the _____; f(−x) is a reflection in the _____.

a · f(x) with a > 1 is a vertical _____; with 0 < a < 1 it is a vertical _____.

Part 3 · Examples (work them with the class)

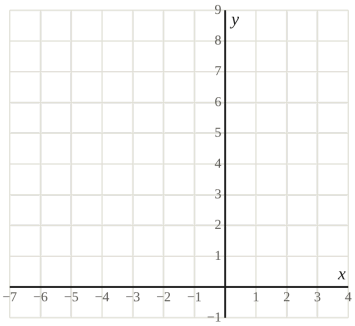
Example 1

Identify the function family of each function.

a. f(x) = |x − 1| b. g(x) = −3x² + 5 c. h(x) = 2x − 7 d. k(x) = 6

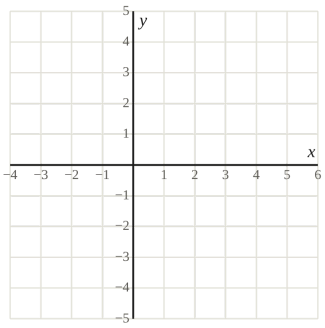
Example 2

Describe the transformation of $f(x) = x^2$ represented by $g(x) = (x + 3)^2$. Then graph both.



Example 3

Describe the transformation of $f(x) = |x|$ represented by $g(x) = -2|x - 1| + 3$.



Example 4

Give the function family, domain, and range of $f(x) = (x - 2)^2 - 5$.

Part 4 · Partner practice

Partner 1

Describe the transformation of $f(x) = x^2$ represented by $g(x) = \frac{1}{2}x^2$.

Partner 2

Describe the transformation of $f(x) = |x|$ represented by $g(x) = -|x + 4| - 2$. Give the range.